

Curriculum Vitae: Dr. Swapan Mallick

Senior Research Scientist,
SMHI / Swedish Meteorological and Hydrological Institute
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RESEARCH SPECIALIZATION AND INTEREST:

- Data assimilation (EnKF, 3D/ 4D-VAR, Hybrid) using GSI-EnKF, WRF, DART, HARMONIE-Arome.
- Numerical weather prediction using WRF, UM (UKMO), HiRLAM (ACCORD, ECMWF).
- Severe weather and impact-based weather prediction and impact study.
- Artificial intelligence, deep Learning (AI/ DL) , Diffusion Model, Convolution NN.
- Data QC, Thinning, Analysis, and Assimilation with Satellite, RADAR, and conventional data.
- GOES-R all-sky radiances, CWP, AMVs, and GLM lightning analysis and assimilation.
- Hyperspectral radiances (from AIRS, IASI, CrIS) assimilation.
- Ensemble forecasts, operational real-time assimilation, observation impact study
- Coupled modeling and Data Assimilation.

RESEARCH PUBLICATION:

- 1 **Swapan Mallick**; Guedj, S.; Lindskog, M. Improving Land Surface Emissivity for Better Simulation of Microwave Radiances over Northern Latitudes. *Remote Sens.* **2026**, *18*, 2819. <https://doi.org/10.3390/rs18162819>.
- 2 **Swapan Mallick**, J. Bojarova, H. Schyberg, P. Dahlgren, E. Støylen, X. Yang, and K. Tølløse, 2026: Copernicus pan-Arctic regional reanalysis (CARRA2): Uncertainty estimation method. Ecmwf report, ECMWF. URL <https://confluence.ecmwf.int/display/CKB/Copernicus+pan-Arctic+Regional+Reanalysis+%28CARRA2%29%3A+Uncertainty+estimation+method>.
- 3 Lindskog, M., Mile, M., Mats Dahlbom, Eriksson, P., Guedj, S., Per, D., Schönoch, D., Hagelin,, **Mallick**, S., Tonboe, R. T., Azad, R., Barlakas, V., Hanna, H., O'Connor, E., Bjarne, A., Eresmaa, R., von Lerber, A., Tyynelä, J., & Dybbroe, A. (2025). Technical note describing the assimilation impact (Version v1). Zenodo. <https://doi.org/10.5281/zenodo.15569907>.
- 4 Schyberg, H., et al. (2025). *Copernicus pan-Arctic Regional Reanalysis (CARRA2): Full system documentation*. Technical report, Danish Meteorological Institute (DMI) and partners. [Available at ECMWF Confluence](#).
- 5 Yang, X., et al. (2025). *Copernicus pan-Arctic Regional Reanalysis (CARRA2): Test and evaluation report*. Technical report. [Available at ECMWF Confluence](#).

- 6 **Swapan Mallick** 2022: *Impact of Adaptively Thinned GOES-16 Cloud Water Path in an Ensemble Data Assimilation System*. *Meteorology* 1, no. 4: 513-530. <https://doi.org/10.3390/meteorology1040032>.
- 7 **Swapan Mallick** and Thomas A. Jones 2022: *Impact of Adaptively Thinned GOES-16 All-sky Radiances in an Ensemble Kalman Filter based WoFS*. *Atmospheric Research*. <https://doi.org/10.1016/j.atmosres.2022.106304>.
- 8 **Swapan Mallick** and Thomas A. Jones 2020: *Assimilation of GOES-16 Satellite derived Winds into the Warn-on-Forecast System*, *Atmospheric Research*, Volume 245, doi: <https://doi.org/10.1016/j.atmosres.2020.105131>.
- 9 **Swapan Mallick**, D. Dutta, and K.-H. Min 2017: *Quality assessment and forecast sensitivity of global remote sensing observations*. *Adv. Atmos. Sci.*, 34 (3), 371-382 doi:10.1007/s00376-016-6109-8.
- 10 D Dutta, AJ Kasimahanthi, **Swapan Mallick**, JP George, PK Devarajan 2017: *Quality assessment of VVP winds from Indian Doppler weather radars: a data assimilation perspective*, *Journal of Applied Remote Sensing* 11 (3), 036021, doi: 10.1117/1.JRS.11.036021.
- 11 C.K. Unnikrishnan, John P. G., A. Lodh, D.K. Maurya, **Swapan Mallick**, E.N Rajagopal and Saji Mohandas, 2016: "Validation of two gridded soil moisture products over India with in-situ observations", *J. Earth. Syst. Sci.*, (Springer), 125(5), pp 935–944. doi 10.1007/s12040-016-0714-x.
- 12 D. Srinivas, Indira Rani S., **Swapan Mallick**, John P. George, 2016: *Impact of AIRS radiance in the NCUM 4D-VAR assimilation system*. *Proc. SPIE* 9880, Multispectral, Hyperspectral, and Ultraspectral Remote Sensing Technology, Techniques and Applications VI, 98800O; <http://dx.doi.org/10.1117/12.2223516>
- 13 **Swapan Mallick**, Indira Rani S., D. Srinivas, John P. George, 2016: *Assimilation of CrIS hyperspectral radiances in a 4D-VAR assimilation system*. *Proc. SPIE* 9880, Multispectral, Hyperspectral, and Ultraspectral Remote Sensing Technology, Techniques and Applications VI, 98800Q; doi: 10.1117/12.2222775; <http://dx.doi.org/10.1117/12.2222775>
- 14 Devajyoti Dutta, **Swapan Mallick**, K. A. Jyothi, John. P. George and D. Preveen. Kumar, 2016: *Wind from Indian doppler weather radars: a data assimilation viewpoint*. *Proc. SPIE* 9876, Remote Sensing of the Atmosphere, Clouds, and Precipitation VI, 98760W; doi:10.1117/12.2223691
- 15 Priti Sharma, Indira Rani S., **Swapan Mallick**, Desamsetti Srinivas, John P. George, Munmun D. Gupta, 2016: *IASI hyperspectral radiances in the NCMRWF 4D-VAR assimilation system: OSE*. *Proc. SPIE* 9880, Multispectral, Hyperspectral, and Ultraspectral Remote Sensing Technology, Techniques and Applications VI, 98800P; doi:10.1117/12.2225868.
- 16 Vivek Singh, Ashish Routray, **Swapan Mallick**, John P. George, E. N. Rajagopal, 2016: *An effort to improve track and intensity prediction of tropical cyclones through vortex initialization in NCUM-global model*. *Proc. SPIE* 9882, Remote Sensing and Modeling of the Atmosphere, Oceans, and Interactions VI, 988210; <http://dx.doi.org/10.1117/12.2223431>
- 17 S. Indira Rani, Desamsetti Srinivas, **Swapan Mallick**, John P. George, 2016: *Impact of advanced technology microwave sounder (ATMS) data in the NCMRWF 4D-VAR data assimilation system*. *Proc.*

SPIE 9882, Remote Sensing and Modeling of the Atmosphere, Oceans, and Interactions VI, 98820F;
<http://dx.doi.org/10.1117/12.2223501>

- 18 John P George, S. Indira Rani, A. Jayakumar, Saji Mohandas, **Swapan Mallick**, A. Lodh, R. Rakhi, M. N. R. Sreevathsa and E. N. Rajagopal: 2016, "NCUM Data Assimilation System". NCMRWF Technical Report No NMRF/TR/01/2016: Available at <http://www.ncmrwf.gov.in/NCUM-Data%20Assimilation.pdf>
- 19 Srinivas Desamsetti, S. Indira Rani, **Swapan Mallick**, Munmum Das Gupta, John P. George, and E. N. Rajagopal: 2016, "Comparison of NCMRWF and ECMWF Archives of Conventional Meteorological Observations". NMRF/RR/03/2016. Available at http://www.ncmrwf.gov.in/NMRF_RR3_2016.pdf
- 20 Jemma Davie and **Swapan Mallick**, 2015: "Comparing Surface and Upper Air Data against Model Background for the Indian Region". United Kingdom Met-Office (UKMO) Research Report. Available at <http://www.metoffice.gov.uk/media/pdf/p/e/FRT602.pdf>
- 21 Prashant Goswami, **Swapan Mallick**, K. C. Gouda, 2011: *Objective Debiasing for Improved Forecasting of Tropical Cyclone Intensity with a Global Circulation Model*. Mon. Wea. Rev., **139**, 2471–2487. doi: <http://dx.doi.org/10.1175/2011MWR3600.1>
- 22 Prashant Goswami, **Swapan Mallick**, 2011: *Objective Bias Correction for Improved Skill in Forecasting Diurnal Cycles of Temperature over Multiple Locations: The Summer Case*. Wea. Forecasting, **26**, 26–43. doi: <http://dx.doi.org/10.1175/2010WAF2222407.1>

INTERNATIONAL PROJECT WORK INVOLVED:

1. Copernicus Arctic Regional Reanalysis Second Generation (CARRA-2) from September 2023 onward.
2. The Weather Generator Machine Learning Earth System Model, EU and ECMWF, will continue from September 2025 onward.
3. European Space Agency (ESA) Project – Performance Evaluation of Arctic Weather Satellite (AWS) Data Processing and Assimilation, April 2025 – ongoing.
4. On-Demand Extremes Digital Twin, Destination Earth (DEODE)– ECMWF: continuing from April 2024.

WORK EXPERIENCE:

09/2023 -Present

Senior Research Scientist, SMHI, SWEDEN

- Experience includes data assimilation and observation processing using the ECMWF, A consortium for Convective-scale modeling Research and Development (ACCORD) assimilation system.
- Expertise in high-impact severe weather nowcasting and short-range forecasting with the HIRLAM system. Principal contributor to the Copernicus Arctic Regional Reanalysis (CARRA-2) product at ECMWF.
- Key team member and primary contributor to the ECMWF Destination Earth on-demand Extremes (DEODE) phase-2 project.

04/2022-08/2023

Research Scientist, CMCC, ITALY

- Satellite ensemble data assimilation using DART-CAM system at Atmospheric Data Assimilation Group, Climate Simulation and Prediction Division at Euro-Mediterranean Centre for Climate Change (CMCC) a private company in Bologna, Italy.

02/2017 – 01/2022

Post-Doc Research Associate, NOAA-NSSL, CIWRO, USA

- Worked on NOAA-NSSL WoFS system, data assimilation using GSI-EnKF and WRF model. GOES-R Data analysis and assimilation including clear and all-sky radiances, Cloud Water Path, and Atmospheric Motion Vector, Lightning Data with WRF and GSI-EnKF system operationally and research mode.
- Hyperspectral Radiance data from AIRS, CrIS and IASI analysis and assimilation and the impact study in convective scale system GSI-EnKF and WRF ensemble bases WoFS system.
- Using convolution neural network (DL) for cloud pattern classification from satellite images and application of AL/ML/DL for convective scale forecasts.

01/2016 – 01/2017

Post-Doc Research Associate, KN University, SOUTH KOREA

- Convective permit variational data assimilation with radar and satellite data. Impact of observation study to improve the forecast skill using conventional and satellite data assimilation. High-resolution KMA UM model and WRF model to assimilate radar reflectivity and in-situ observation. Worked at Kyungpook National University, Daegu, South Korea.

10/2012 – 01/2016

Project Scientist, NCMRWF, MoES, INDIA

- Numerical Modelling of Weather and Climate and Indian Monsoon Data Assimilation and Analysis (IMDAA) at NCMRWF, India (with Met-Office, UK).
- Global Numerical Weather Prediction (UKMO Unified Modelling system), Variational Data Assimilation (Satellite and conventional data), and Observation Processing System (OPS) using global Unified Model (Met-Office, UK). Results are available in publication list and the NCMRWF publication/ report list [www.ncmrwf.gov.in]

12/2005 – 10/2012

Research Fellow, CSIR CMMACS, INDIA

- Numerical Weather Prediction (LMDz GCM France, WRF). Developed Non-linear Algorithms for Model Bias Correction.
- Variational Assimilation (3D, 4D-VAR). Tropical Cyclone Track, Intensity and genesis forecast using GCM. Model Output Statistics, Station Scale forecast using NCEP Reanalysis data. Station scale forecast

using global Circulation model GCM and analysis with CSIR tower data (CoMMON).

EDUCATION:

- 2007 - 2013** Ph.D. in Physics (Data Assimilation, NWP), CSIR C-MMACS, Mangalore University
- 2000 - 2002** M.Sc. in Applied Mathematics, Gauhati University
- 1997 - 2000** B.Sc. in Mathematics, Assam University

CONFERENCE PRESENTATION, REPORTS (Ten Last):

1. **Swapan Mallick**, "Derivation of Structure Functions over the Copernicus Arctic Regional Reanalysis (CARRA-2) Region, A consortium for Convective-scale modeling Research and Development (ACCORD), 4th All Staff Workshop, 15-19-April-2024, Norrkoping, Sweden. Available at https://www.umr-cnrm.fr/accord/IMG/pdf/swapan_structurefunction_accord.pdf
2. **Swapan Mallick**, Jelena Bojarova, Xiaohua Yang, "High-resolution Data Assimilation and the Structure Function for the Copernicus Arctic Regional Second Generation Reanalysis" Jun. 7, 2024, Convective-Scale Data Assimilation II, ISDA available at https://www.data-assimilation.riken.jp/en/events/isda_2021/index.html
3. **Swapan Mallick** and Thomas A. Jones, "Impact of Adaptively Thinned GOES-16 All-sky Radiances and Retrievals Cloud Properties in an Ensemble Kalman Filter based WoFS, January 2022, at 102 American Meteorological Society, Annual Virtual Meeting.
4. **Swapan Mallick**, Thomas A. Jones, Kristopher Bedka, William Smith Jr., and Rabindra Palikonda, "Impact of GOES-R Nighttime Cloud Water Path Assimilation into the Warn-on-Forecast System" January 2021, at 101 American Meteorological Society, Annual Virtual Meeting.
5. **Swapan Mallick**, T. A. Jones, K. H. Knopfmeier, P. Skinner, and D. C. Dowell "Quality Assessment and Impact of High-Resolution GOES-16 AMVs into the GSI-EnKF-Based WoFS," January 2020, at 100 American Meteorological Society, Annual Meeting, Boston, MA. Recorded Presentation available at <https://ams.confex.com/ams/2020Annual/videogateway.cgi/id/516591?recordingid=51659>.
6. Thomas A. Jones (presenter), **Swapan Mallick**, K. H. Knopfmeier, D. C. Dowell, X. Wang, P. S. Skinner, P. Minnis, R. Palikonda, and W. L. Smith Jr., 2019: "Status of Satellite Data Assimilation in the NSSL Experimental Warn-on-Forecast System". January 2019, American Meteorological Society, Annual Meeting,
7. Kent Knopfmeier, Nusrat Yussouf, Patrick Skinner, Thomas Jones, Jessica Choate, Katie Wilson, Junjun Hu, **Swapan Mallick**, Anthony Reinhart, and Gerald Creager (CIMMS at NOAA-NSSL)

- 2018: "Storm-Scale Data Assimilation and Ensemble Forecasting with the NEWS-e", Cooperative Institute for Mesoscale Meteorological Studies (CIMMS) Annual Report. PP 150-151, Available at https://cimms.ou.edu/largefiles/cimmsfy18_reduced.pdf.
8. **Swapan Mallick** and T. A. Jones, "Satellite Data Assimilation into the High-Resolution WoF GSI-EnKF system" Convective meteorology Seminar at School of Meteorology, The University of Oklahoma, Norman on 2nd November 2018.
 9. **Swapan Mallick** and Thomas Jones, 2018 "Storm-Scale Assimilation of Hyperspectral Infrared Clear-Sky Radiances" Cooperative Institute for Mesoscale Meteorological Studies (CIMMS, NOAA-NSSL) Annual Report. Available at https://cimms.ou.edu/largefiles/cimmsfy18_reduced.pdf.
 - 10.

TECHNICAL SKILLS:

1. Software and programming languages: FORTRAN, Python, GitHub, SQL, Unix shell scripting, TensorFlow, and PyTorch for AI, machine learning, and deep learning.
2. AI/ML Systems: Machine Learning, Deep Learning, Diffusion Models, and Convolutional Neural Networks.
3. Data assimilation using high-resolution GSI-EnKF with WRF, JEDI with FV3, and 4D-VAR with the UK Met Office Unified Model.
4. In-depth knowledge of FORTRAN programming, with extensive experience developing FORTRAN-90/95 codes for both operational and research purposes.
5. High-Performance Computing, Numerical Weather Prediction using WRF, the UK Met Office Unified Model, and the France LMDz General Circulation Model (GCM).
6. Large datasets from satellites, reanalysis, and conventional sources are available in various formats, including NetCDF, ZARR, BUFR, OBSTORE, HDF5, and GRIB.
7. Experience working with RTTOV and CRTM to generate synthetic satellite data.

CONFERENCE / WORKSHOP/ TRAINING:

1. "7th JEDI (The joint effort of data assimilation integration) academy with FV3". October 4-8, 2021, held virtually.
2. "Unified Forecast System (UFS) for Short-Range Weather Application" workshop and short course organizes by the Developmental Testbed Center (DTC), UCAR on 20-24 September, 2021.
3. "Machine Learning in Python for Environmental Science Problems" short course organizes by American Meteorological Society (AMS) on 8-9 April 8-9, 2021.
4. "Introduction to the Joint Effort for Data Assimilation Integration (JEDI)" short course organizes by American Meteorological Society (AMS) on March 10, 2021.
5. "GOES-R/JPSS Hands-on Training to Process, Display, and Analyze Satellite Data Products" short course organizes by American Meteorological Society (AMS) on March 17 and 18, 2021.
6. "Introduction to the WRF-Hydro Modeling System" short course organizes by American Meteorological Society (AMS) on 23 February 2021.
7. Participate in NOAA's 2nd Virtual Workshop on "Leveraging Artificial Intelligent (AI) in Environmental Sciences". From 30 July 2020 to 25 Feb 2021.
8. "Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning" April 2020, an online non-credit course authorized by Coursera.

9. Participated in NOAA's 1st Workshop on *"Leveraging AI in the Exploitation of Satellite Earth Observations & Numerical Weather Prediction"* during April 23-25, 2019 at NOAA Center for Weather and Climate Prediction (NCWCP), Maryland.
10. "Neural Networks and Deep Learning", May 2019, an online non-credit course authorized by *deeplearning.ai* and offered through *Coursera*.
11. "Satellite Data Processing and Fast Radiative Transfer Modelling", 5 days training course at NCMRWF, Course conducted by Dr. Roger Saunders, Met Office, UK during 23rd March to 27th March 2015.
12. "Short term training course in Operational Meteorology and Indian Summer Monsoon" at India Meteorological Department (IMD), New Delhi; 2nd-20th September 2013.
13. "Short course on Ensemble Kalman Filtering - Methods and Algorithms" Algorithms during 15th-26th July 2013. Course conducted by Prof. S. Lakshmi Varahan, School of Computer Science, University of Oklahoma, at ITCOO-INCOIS, Hyderabad, INDIA.
14. "Data Assimilation and use of Satellite Data" from 3rd-12th, June 2013 at European Centre for Medium Range Weather Forecasts (ECMWF), Reading, United Kingdom (UK).
15. "Summer School on Data Assimilation" from 17th-21st December 2012 conducted by Indian Institute of Remote Sensing (IIRS, India), University of Reading (United Kingdom UK) and the UK Space Agency at IIRS, India.
16. "Data Assimilation Course on Global Model" at National Centre for Medium Range Weather Forecasting (NCMRWF), Noida, India, from 23rd-27th November 2009.
17. "Intensive Course on Inverse Modelling", course conducted by Prof. Albert Tarantola, Institut de Physique du Globe de Paris, organized at CSIR Centre for Mathematical Modelling and Computer Simulation (C-MMACS), Bangalore, from 19th-27th June 2006.

PERSONAL DETAILS:

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